Weather Classification with CNNs: Successes, Surprises, and Sunrise Confusion

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ABSTRACT

- Trained a CNN on 1500 labeled weather images from Kaggle.
- Classified images into five weather types.
- Assessed performance using accuracy and confusion matrices.

INRODUCTION

- Weather image classification aids meteorology and automation.
- **Challenges**: data imbalance and varying image sizes.
- Used CNNs with Python tools for classification.

METHODS

- **Split data:** 85% train, 15% validate across five classes.
- Used Conv2D, Pooling, Flatten, Dense, Dropout layers.
- **Augmentation**: rescale, rotate, zoom, and flip.

RESULTS

- Accuracy improved steadily with no overfitting.
- Training and validation were stable.
- Misclassification mainly affected 'sunrise'.

CONCLUSION

- CNNs can classify weather from photos effectively.
- Augmentation boosts model accuracy.
- More work needed on weaker classes like 'sunrise'.

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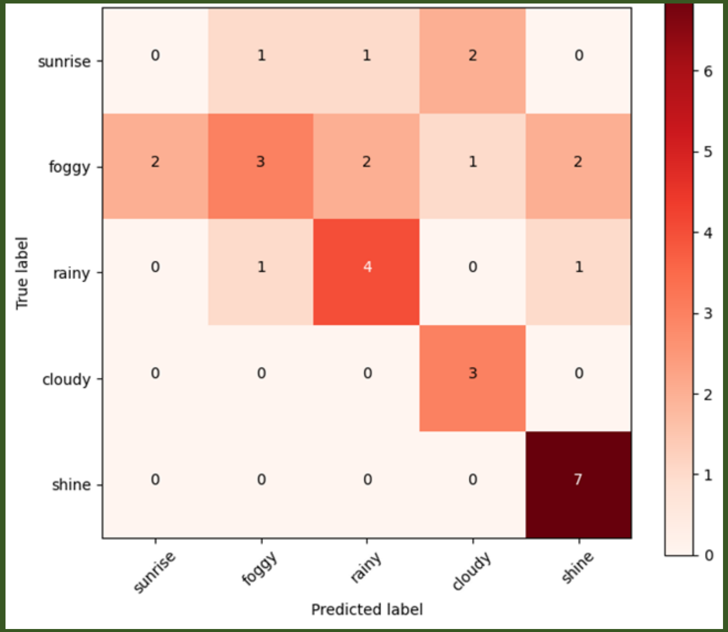
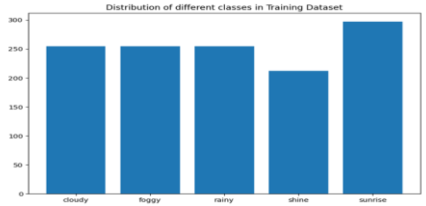
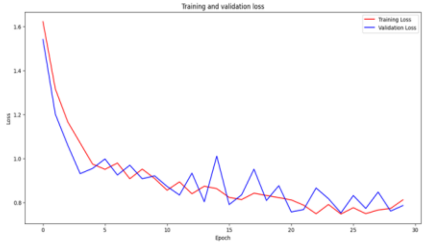
A computer model trained to identify weather conditions from photos, performed well overall, but still struggles to recognize sunrise images.



OTHER GRAPHS





LIMITATIONS

- Poor performance on 'sunrise' class.
- Dataset may be too small for generalization.
- Image quality and imbalance affected results.

FURTHER RESEARCH

- Use larger and more diverse datasets.
- Try fine-tuning or transfer learning.
- Test on real-world weather images.

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